Introduction to AVR Microprocessor Development Board

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This document gives you a brief description on the Microchip Microcontroller Development Board and how to use it.

1. Equipment Handling Precaution

A few of precautions are necessary when using the board:

- Your experiment place, either laboratory or home, can be an environment where it is quite easy for you to build up a static electric charge. **Electrostatic discharge can destroy electronic equipment**.

  Since you may be carrying a static charge without even realizing it, **you should always discharge yourself**. You can do this painlessly by touching a grounded conductor using a coin, a key or other metallic objects instead of your finger.

- Short circuits may damage certain devices. Please remove any metal items before handling the board. There are no dangerous voltages or currents that will harm you here, but this is a good habit to always act with caution and follow the designed procedure.

- Always turn the power off before connecting or disconnecting any I/O subsystems and don't place the board on a conductive surface when the power is on.

2. Experiment Setup

Figure 1 shows the lab board that contains peripherals, connectors, and the Arduino MEGA 2560 R3 microcontroller (at the back).

To program the board, you need first to download and install the following software (available on the course website on the References page)

- arduino-1.0.6-windows.exe

  (Ardiuno: Open-source electronic prototyping platform enabling users to create interactive electronic objects.)

To connect the board to your computer, use the USB Type-B port on the Arduino (left side). When using the lab computers, use the provided USB cable instead of pulling your own one into the computer.

The communication between the board and the computer requires a correct COM port setting. To work out which COM port the Arduino is connected to, open the system properties of the computer by clicking the start button, right-clicking 'My Computer' then clicking properties. In the 'Hardware' tag, click 'Device Manager'. Scroll down to 'Ports (COM & LPT)'. There should be one item called 'Arduino Mega 2560 (<comport>)'.

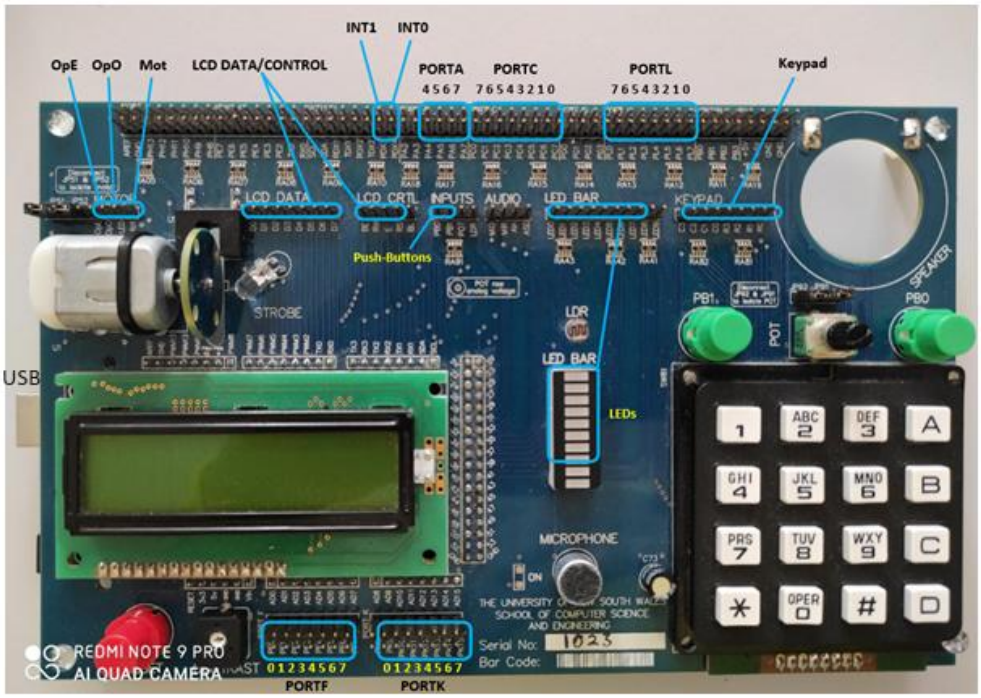


Figure 1: AVR Development Board

3. Download and Run Code on the Board

Below are two ways to download code (test.hex) on the lab board. To execute the code provided (test.hex), connect the LEDs to PORT C (See Figure 1 for the related pin locations).

Method 1: download from the command window

Download 'programmer.zip' from the Labs page under the link of Lab Board Information on the course website (the same location as this document). Extract the files and open the folder.

Run the batch file 'console.bat' to open a command window as shown in Figure 2.

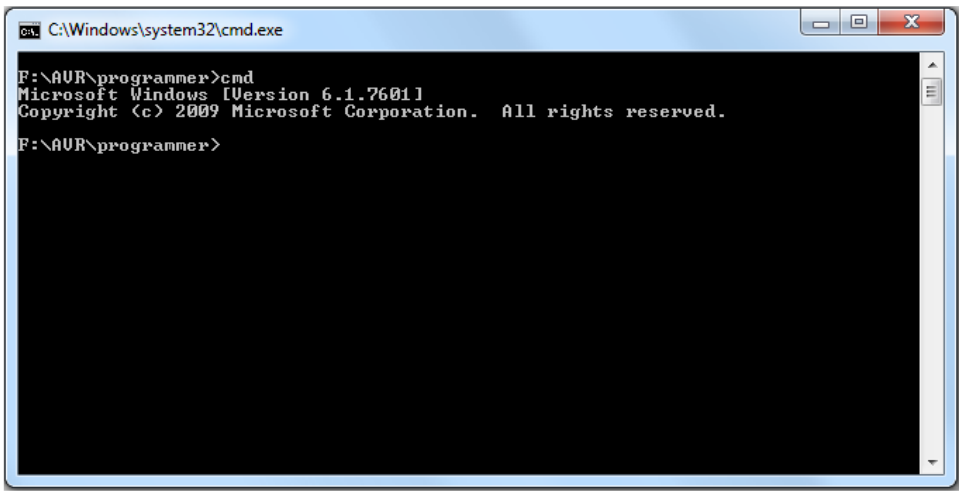


Figure 2

In the window, type the following command to load the test program in the folder with the following download command:

```
download <comport> test.hex
```

Here <comport> is the name of the port you found earlier, e.g. COM5.

A message will be returned to indicate the success of download, as shown in Figure 3.

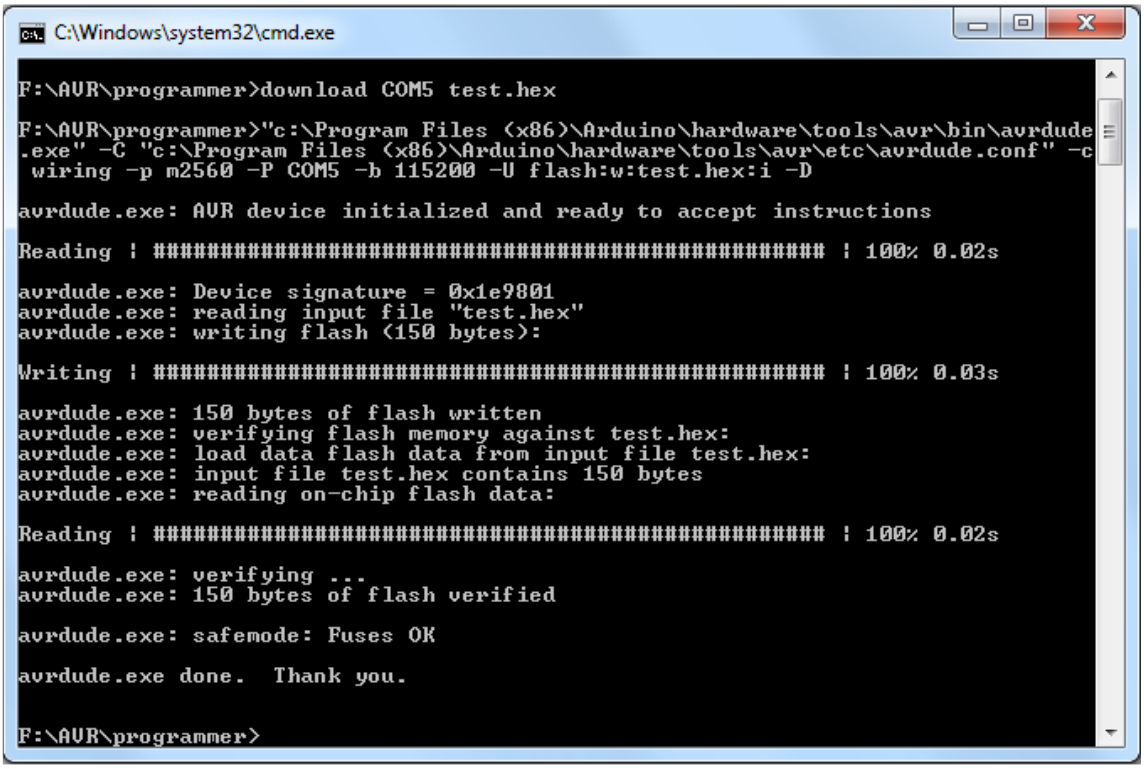


Figure 3

Once the download is complete and the LEDs are connected correctly, the LEDs should start flashing.
You can run your own code generated by the Microchip Studio. To download it, replace ‘test.hex’ with the path to your code. For example,

```
download <comport> F:\AVR\test2.hex
```

Method 2: download from Microchip Studio

1) Go to the menu item *Tools->External Tools* in Microchip Studio

2) Fill in the dialogue box as follows.

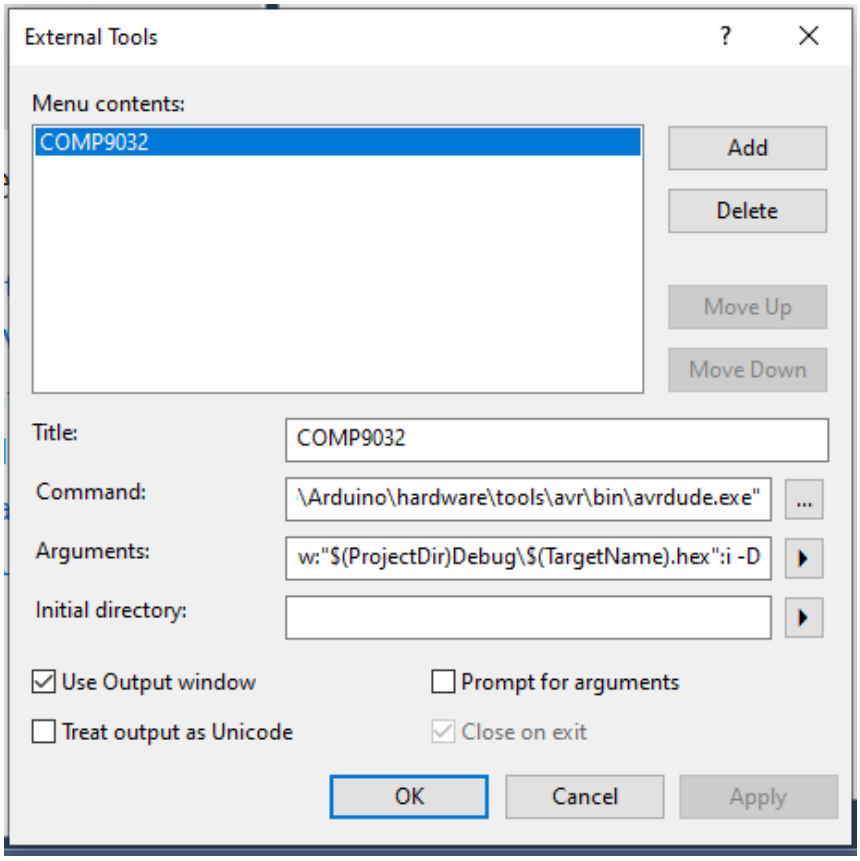


Figure 4

- Give any name you like for the *Title*.
- The *Command* should be the location of the *avrdude* program – the command line program that performs the downloading: *"C:\Program Files (x86)\Arduino\hardware\tools\avr\bin\avrdude.exe"*
- The *Arguments* should be *-C "C:\Program Files (x86)\Arduino\hardware\tools\avr\etc\avrdude.conf" -c wiring -p m2560 -P COMX -b 115200 -U flash:w:"$(ProjectDir)Debug\$(TargetName).hex":i -D*

  where *COMX* is the related COM port you found earlier.

3) Select Use Output window, then click Apply and OK

4) In an opened project, click on the added tool name, as shown in Figure 5. The .hex file generated in the project will be downloaded.

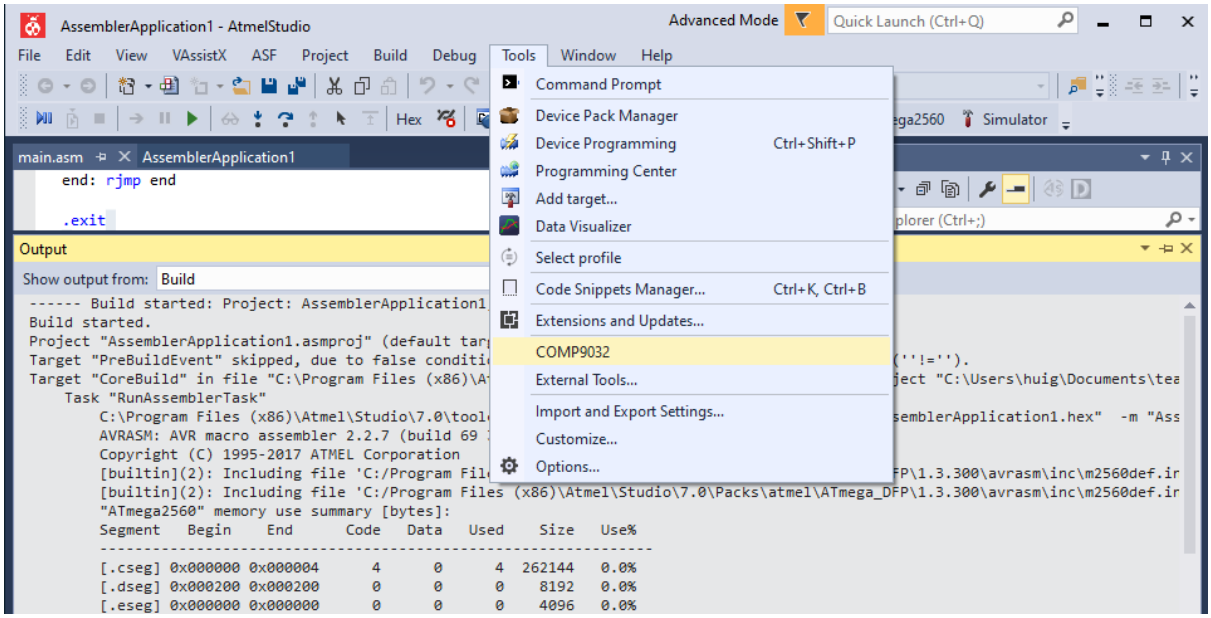


Figure 5

5) You should see the output of the download as shown below.

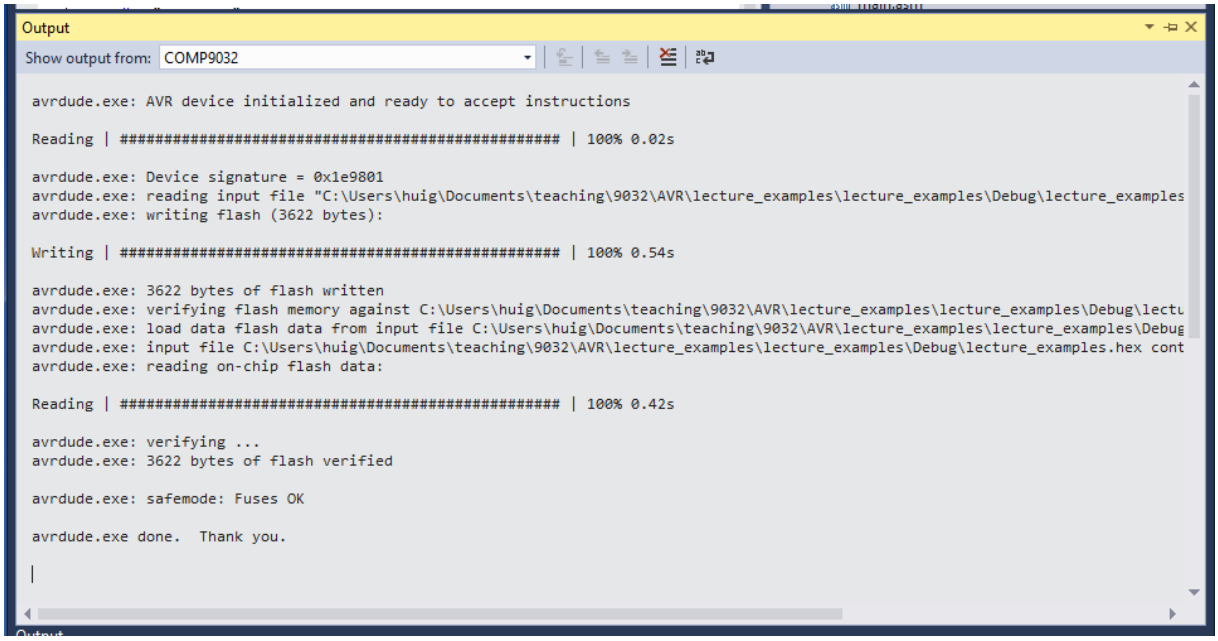


Figure 6

4. Board Test

Before you do experiments with your board, please test its functionality by following the test procedure given in “Board Test Procedure.pdf”.

Refer to the video “Lab board setup” on the same web page for a demonstration.